Language Identification Using Score Vector Modeling and Support Vector Machine

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   - Previous Work
   - Our Motivation
2. Our Approach
   - The Proposed LID System
   - Features
   - Score Vector Modeling
   - Language Score Calibration
3. Experiments and Results
   - Corpora
   - Results on NIST LRE 2003 30 Seconds Task
4. Brief summary

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PPRLM (Phoneme Recognizer Followed by Language Models)

1. PPRLM with phoneme strings
   - *MA Zissman (1995)*
   - *Y Yan and E Barnard (1995)*
2. PPRLM with phoneme lattices
   - *JL Gauvain, A Messaoudi and H Schwenk (2004)*
3. PPRLM with high quality phoneme recognizer
   - *P Matejka, P Schwarz, J Cernocky, P Chytil (2005)*

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GMM (Gaussian Mixture Model) with SDC (shifted-delta-cepstral) Acoustic Features

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2. GMM-MMI
   - *L Burqet, P Metejka, J Cernocky (2006)*

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Phonotactic Modeling Based on Support Vector Machines

1. Score vector modeling with discriminative classifiers
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SVM with SDC Features and GLDS (Generalized Linear Discriminate Sequence) Kernel

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$$\begin{aligned} K(X,Y) &= \frac{1}{N_x}\sum_{i=1}^{N_x} b(x_i)^t \cdot \overline{R}^{-1} \cdot \frac{1}{N_y}\sum_{j=1}^{N_y} b(y_j) \\ &= \overline{b_x}^t \cdot \overline{R}^{-1} \cdot \overline{b_y} \end{aligned}$$

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SVM with Speaker-Specific Anchor Model Representation

1. *E Noor, H Aronowitz (2006)*
   - train GMMs for every speaker in each language
   - backend SVM classifiers

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Questions

- limited computer memory vs large database training
- entire anchor GMMs scoring vs computation complexity

Solutions

- multiple speaker group based GLDS kernel classifiers
- backend SVM classifiers
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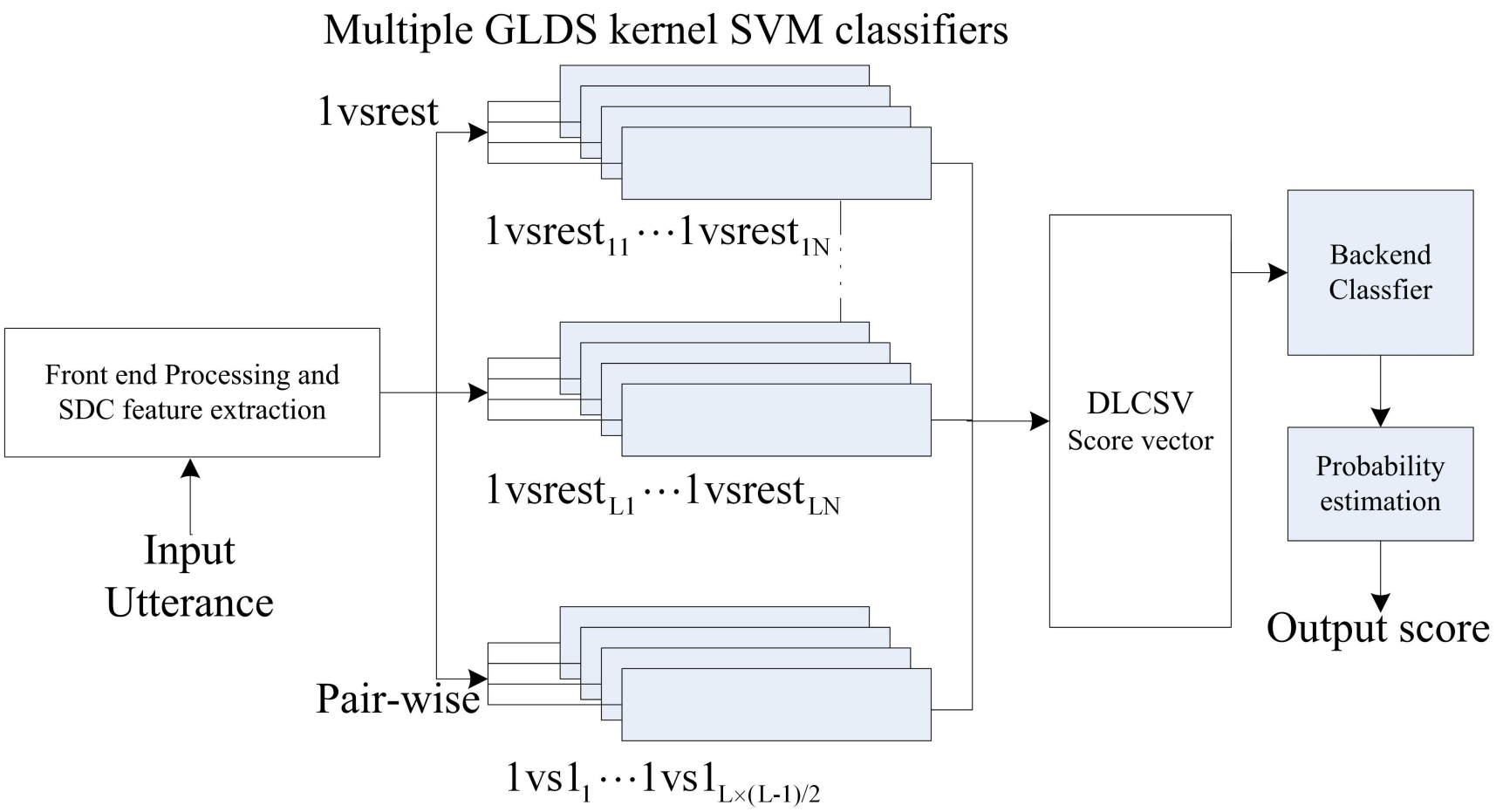
The Proposed LID System



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56 Dimension SDC Features

- 7 MFCC coefficients (including C_0) concatenated with SDC 7-1-3-7 feature —— *L Burqet, P Metejka, J Cernocky (2006)*
- speech activity detection -> 56 dimension SDC extraction -> cepstral mean/variance normalization

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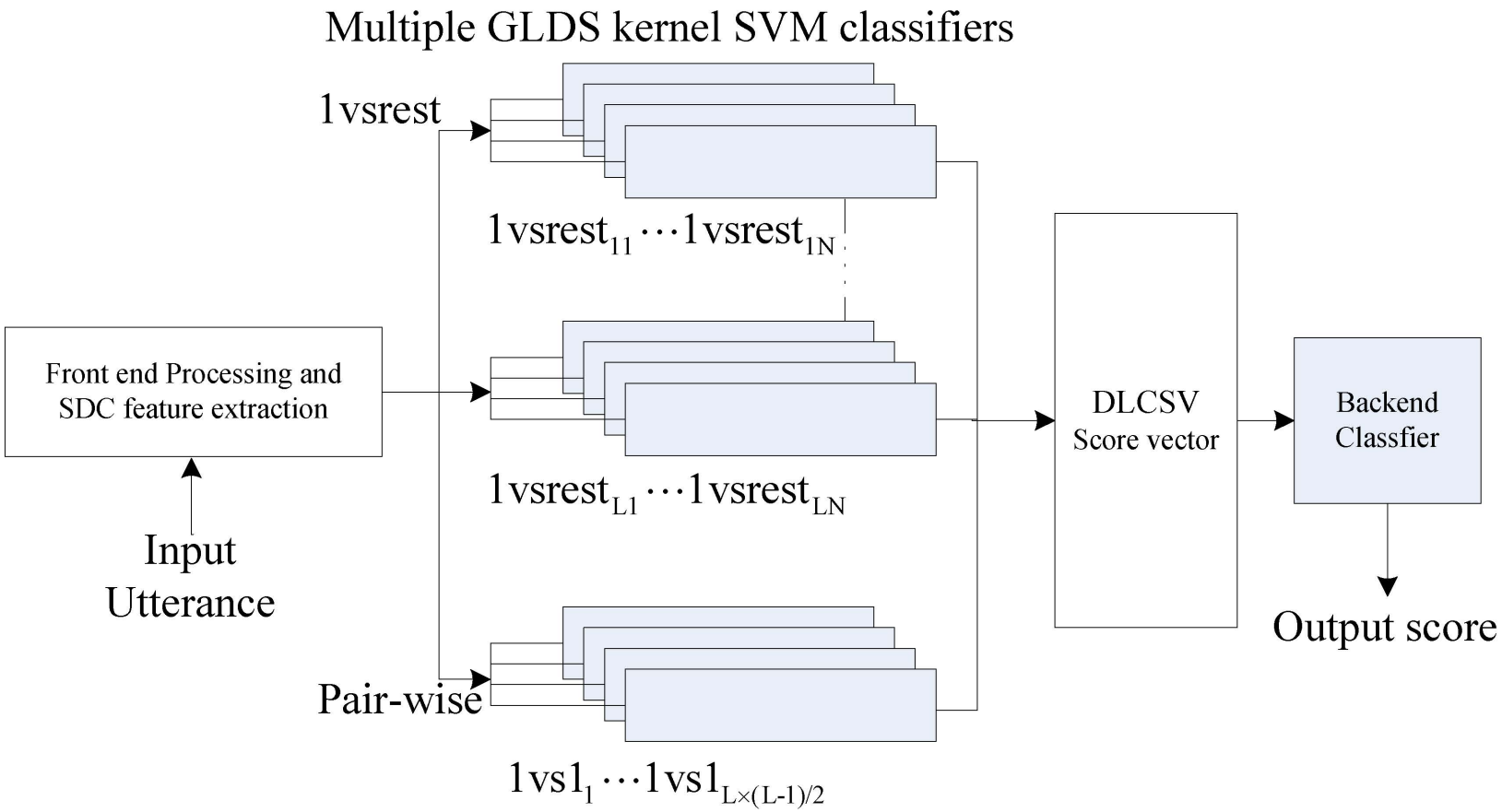
DLCSV (Discriminative Language Characterization Score Vector)



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Language Score Calibration

- log-likelihood ratios (LLR) normalization

$$P_i = \frac{\pi_i e^{S_i}}{\sum_{j=1}^{L} \pi_j e^{S_j}}, i = 1, 2, \cdots, L$$

$$S_i' = S_i - log(\frac{1}{L-1} \sum_{j \neq i} e^{S_j})$$

- Multi-class Probability Estimates by Pairwise Coupling
  T-F Wu, C-J Lin, RC Weng (2006)

$$r_{ij} = p(y = i | y = iorj, x)$$

$$min \frac{1}{2} \sum_{i=1}^{L} \sum_{j, j \neq i} (r_{ji} p_i - r_{ij} p_j)^2 \quad with \sum_{i=1}^{L} p_i = 1, p_i \geq 0$$

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Corpora

1. Training data for multiple GLDS kernel SVM classifiers
   - CallFriend corpus (train, development and test sets)
2. Training data for backend SVM classifiers
   - NIST 1996 LRE development set
3. Testing data
   - NIST 2003 LRE evaluation set

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SVM System Results on NIST LRE 2003 30 Seconds Task

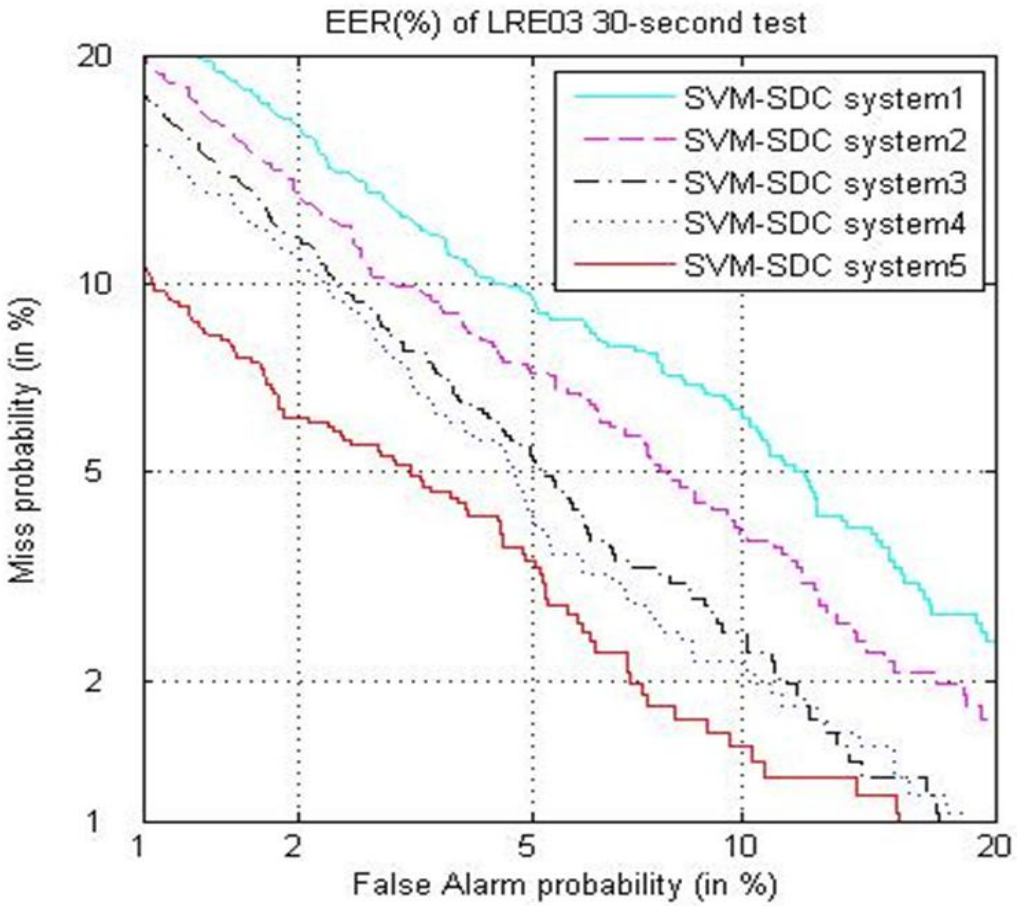
SVM-SDC system	**1**	**2**	**3**	**4**	**5**
DLCSV-12	√				
DLCSV-78		√			
DLCSV-138			√	√	√
LLR normalization	√	√	√		
Probability estimation				√	√
SDC (7-1-3-7)	√	√	√	√	
modified SDC (7-1-3-7 + MFCC)					√
EER-NIST-LRE-03-30s task (%)	7.0	5.9	5.0	4.7	4.0

EER Comparison with State-of-Art SVM System

EER Results on NIST LRE 2003 30 Seconds Task

GLDS SVM	6.1%
Anchor GMM	4.8%
Proposed SVM-SDC system 5	4.0%

DET Curves on NIST LRE 2003 30 Seconds Task



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Thanks!
Questions and Answers